Cheatsheet AZ-305 — 07 · Monitoring

1. Who does what

Service / feature	What it is	Signal in the wording
Azure Monitor	Umbrella platform: metrics, logs, alerts, workbooks	generic
Log Analytics workspace	Log repository queryable with **KQL**; correlates different sources; basis for Log-type alerts	"correlate resource data with application data", "query logs", "alerts from logs"
Application Insights	APM: requests, dependencies, exceptions, **availability tests** (synthetic, from the Internet), Application Map, response-time alerts	"simulate user connections/transactions", "response time goals", "performance of the app"
Activity Log	**Control plane** audit (who created/modified/deleted/stopped what)	"who stopped the VM", "resource changes"
Service Map / VM Insights	Topology and dependencies between processes and servers	"dependencies between servers"
Network Watcher	Network diagnostics: IP flow verify, NSG diagnostics, next hop, connection troubleshoot, packet capture, NSG flow logs	"packets allowed or denied", "next hop", "capture traffic"
Workbooks	Interactive reports over logs (e.g. "Inactive user accounts")	"list inactive admins"
Azure Advisor	Recommendations (cost, security, reliability, performance); **does not act**	"recommendations"
Defender for Cloud	Security posture, JIT VM access, threat detection	"secure score", "JIT ports"
Connection Monitor	Latency/availability between endpoints (incl. ExpressRoute)	"connectivity over time"

2. Agents and collection

Element	What it is
Azure Monitor Agent (AMA)	Current agent for Azure VMs and (via **Azure Arc**) on-prem/other clouds. Replaces MMA/OMS
Data Collection Rule (DCR)	What to collect (Windows events, IIS, perf, syslog, custom), from which machines (associations, **any region**), where to (workspace)
Data Collection Endpoint (DCE)	Regional endpoint; **required only with Private Link** or custom/text logs
Azure Arc	Brings external servers in as Azure resources → AMA, Policy, Update Manager
Diagnostic settings	PaaS resources and Entra ID → Log Analytics / Storage / Event Hubs

- **Minimum DCRs:** count **distinct sets of data types**, not regions. Windows logs (Central+East) and IIS logs (East+West) → **2 DCRs**; East has both associated. Workspace over public endpoint = no regional DCE.
- **Centralized** workspace for hybrid monitoring; **resource-context RBAC** so each team sees only its own resources.

3. Private Link for Azure Monitor (AMPLS)

```
Workspace1 ── 1 AMPLS ─┬── Private Endpoint (VNet1 or VNet2, peered → serves both)
                       └── Private Endpoint (VNet3, isolated)
```

- **Private Endpoints:** one per **connectivity group** (peered VNets count as one; isolated ones, one each).
- **AMPLS:** normally **1** — groups several PEs and several workspaces.
- Private DNS zones linked to the VNets to resolve the endpoints.

4. Alerts

Signal type	Evaluates	When
Metric	Platform numeric values (CPU, DTU, requestCharge) in near real time	Simple thresholds
Log	Result of a **KQL query** over Log Analytics	Complex conditions: country, event type, grouped counts, non-compliant Policy, sign-in logs
Activity Log	Management events (power off, restart, deallocate, delete)	Resource changes
Resource Health	Resource availability/health	Platform outages
Service Health	Azure incidents	Maintenance/incidents

- **Action Group:** reusable recipients/actions (email, SMS, webhook, Function, Logic App, ITSM). Several rules share one.
- 3 different VM operations (off/restart/deallocate) → **3 rules + 1 action group**.
- Log alert: **period** (window) and **frequency** (how often) are different from the **threshold**.
- **Autoscale** (VMSS): metric + threshold + **duration** (window) + **cooldown**; **multiple profiles** by schedule/date for seasonal traffic. VMSS scales **horizontally**, never vertically.

5. Log Analytics costs

Lever	What it does	Side effect
Commitment tier (100 GB/day, 200…)	Lower price per GB in exchange for a daily commitment	**None**: everything is ingested, alerts intact
Basic Logs (per-table data plan)	Cheaper ingestion	Limited KQL, **no log search alerts**, short retention
Auxiliary Logs	Even cheaper	Basic queries only
Daily cap	Daily ingestion limit	**Data is lost** once exceeded; alerts go blind
Retention	Interactive (up to 2 years) + archive (up to 12 years)	Retention cost

- "Minimize costs, ingest all logs, don't impact the alert" → **commitment tier** (resource: **Workspace1**; the data plan belongs to the table; daily cap discards data).
- 7-year retention of security logs → workspace with archive or export to Storage.

6. Application Insights vs Log Analytics (the distinction that gets tested)

Question	Answer
Correlate resource usage with application data	**Log Analytics** (common repository, KQL)
Simulate users from the Internet, measure transactions, alert on response time	**Application Insights** (availability tests)
See who did what in the portal	**Activity Log**
Dependencies between servers	**Service Map**

7. Decision tree

```
What is the question?
├─ App performance, transactions, synthetics ..... Application Insights
├─ Query/correlate logs, query-based alerts ..... Log Analytics + Log alert
├─ Management audit, shutdowns, changes ..... Activity Log
├─ Network: allowed/denied, next hop, capture ..... Network Watcher
├─ Recommendations without acting ..... Advisor
├─ Act automatically (SQL) ..... Automatic tuning
├─ Collect from hybrid VMs ..... Arc + AMA + DCR + central workspace
├─ Logs over private backbone ..... AMPLS + Private Endpoints per VNet group
└─ Reduce cost without losing data or alerts ..... Commitment tier
```

8. Signals → answer

Wording	Answer
"correlate … with …"	Log Analytics
"simulate user connections from the Internet"	Application Insights
"sign-in logs" + "store and alert"	Log Analytics + signal type Log
"who powered off the VM"	Activity Log alert
"minimum DCRs"	Sets of data types
"minimum private endpoints / AMPLS"	Connectivity groups / 1
"improve SQL performance, minimize admin effort"	Automatic tuning
"packets allowed or denied"	Network Watcher IP flow verify / NSG diagnostics
"120 GB/day, pay-as-you-go, keep all logs, keep alert"	Commitment tier on the workspace
"Basic Logs" on a table with an alert	Breaks the alert → ruled out
"daily cap" + "ingest all"	Loses data → ruled out
"hybrid servers"	Azure Arc + AMA

9. Mistakes to avoid

1. Application Insights to correlate infrastructure and app → Log Analytics.
2. Metric alert for conditions that need content filtering (country, type) → Log.
3. One DCR per region → count data types.
4. One AMPLS per VNet → one is enough.
5. Basic Logs with active alerts.
6. Vertical scaling on VMSS.